MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

MARCH 2026 SEMESTER

CLOUD COMPUTING (IIB43203)

GROUP PROJECT ASSIGNMENT

PREPARED BY:

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PREPARED FOR:

MEGAT NORULAZMI BIN MEGAT MOHAMED NOOR

SUBMISSION DATE:

Sunday, 7 June 2026

Group Project Assignment

For this group project, you need to form a group of a maximum of four members and a minimum of two members

Deliverables must be:

1. Write a Report and a step-by-step guide for the reader to follow and replicate your project.

2. Create a video presentation about your project (All members must present) and publish the video presentation to your YouTube channel (put the link to the video in your report)

3. Save all your project materials, including the report, into your GitHub account.

Your task for this group project:

Using your AWS trial account, create a web application or a website based on Compute, Storage, Networking, and Database AWS Services

Marking criteria:

- Comprehensive instructional/tutorial guide and project reporting

- Comprehensive video tutorial explanation (ALL Team Members Must Present)

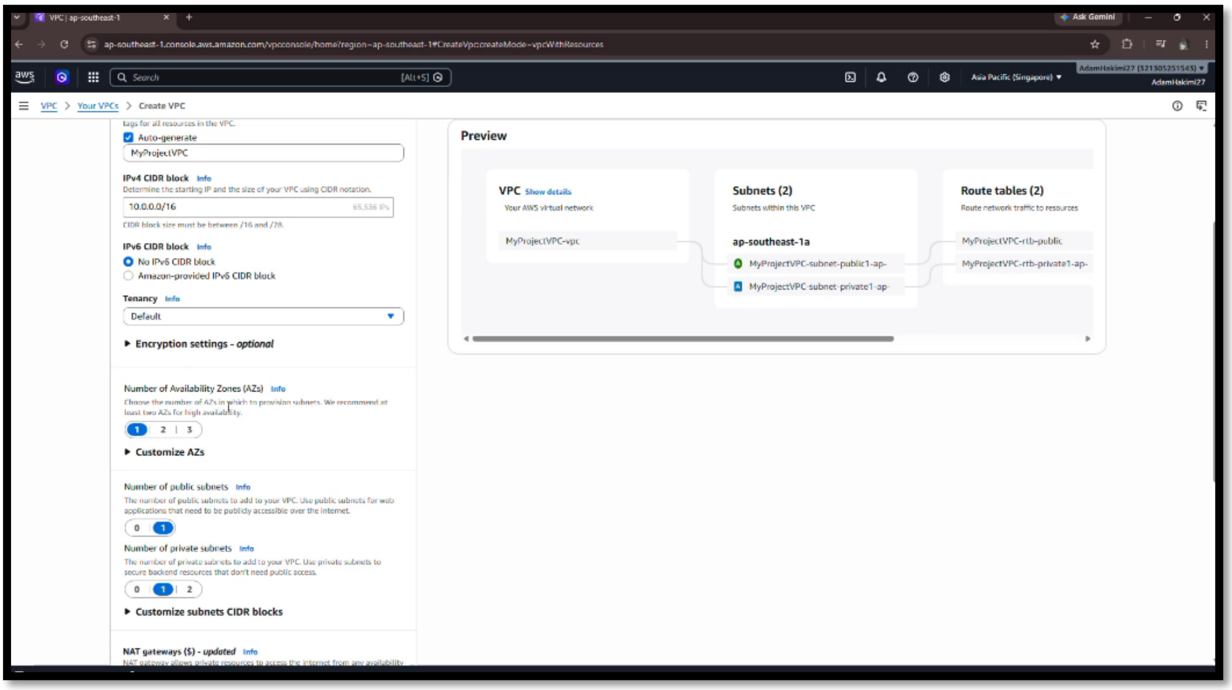
- Good quality presentation materials

- Good Creativity

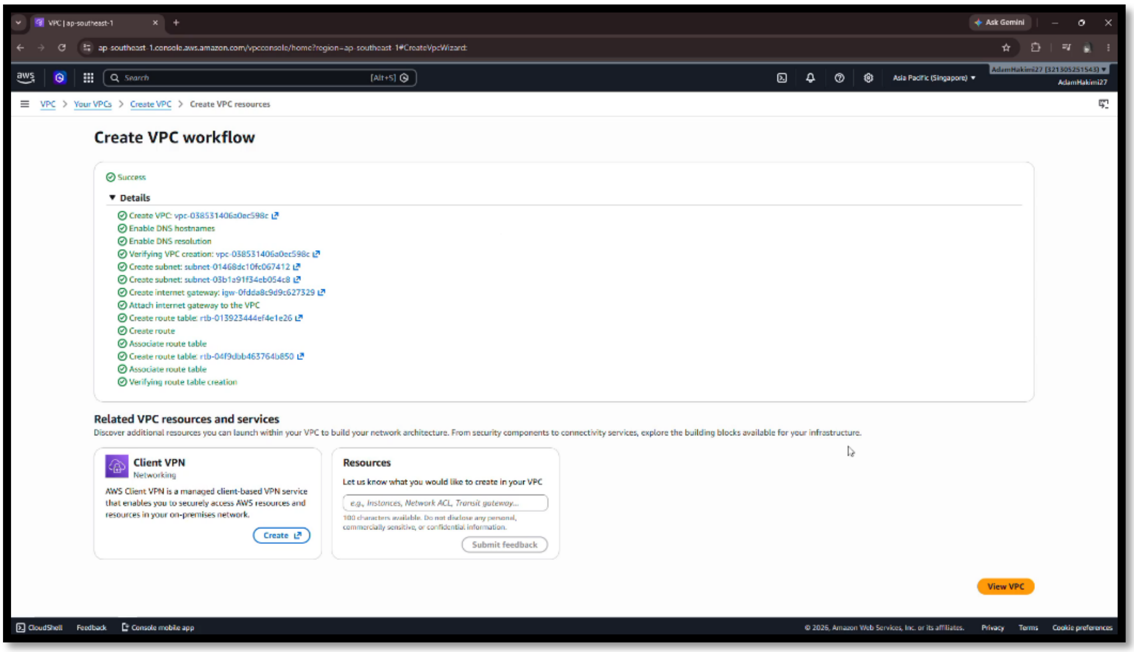
- Show the outcome based on teamwork

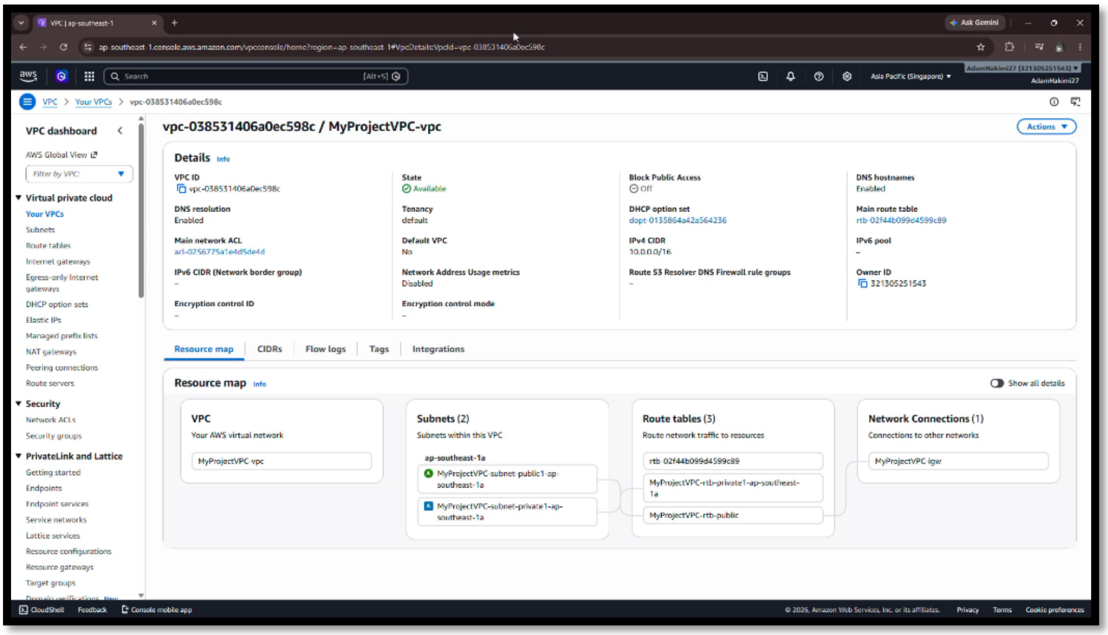
Please start early, and all the best!

Creating VPC



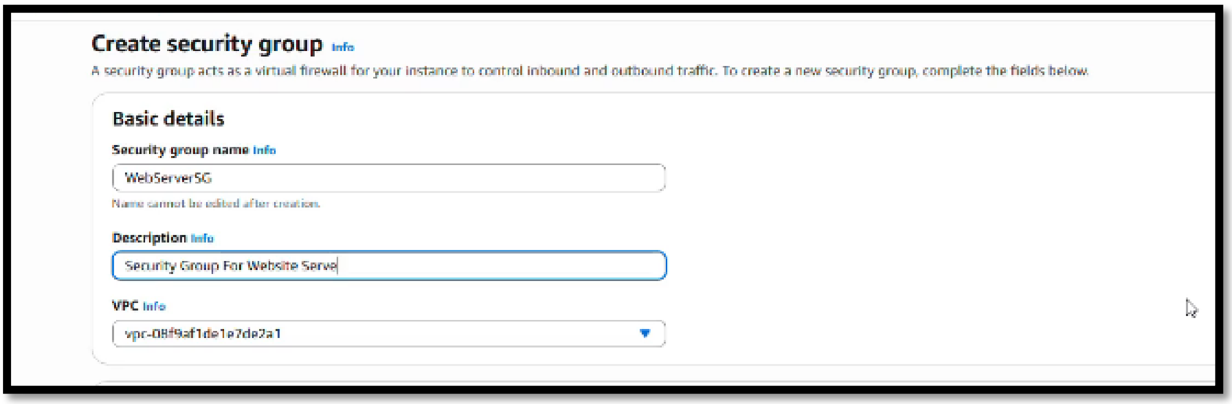
Creating VPC. Renaming the VPC to MyProjectVPC. Keep the IPv4 CIDR block as default. Change the Number of Availability zones to 1 and all the rest will follow to 1 like the number public of subnets and number of private subnets.



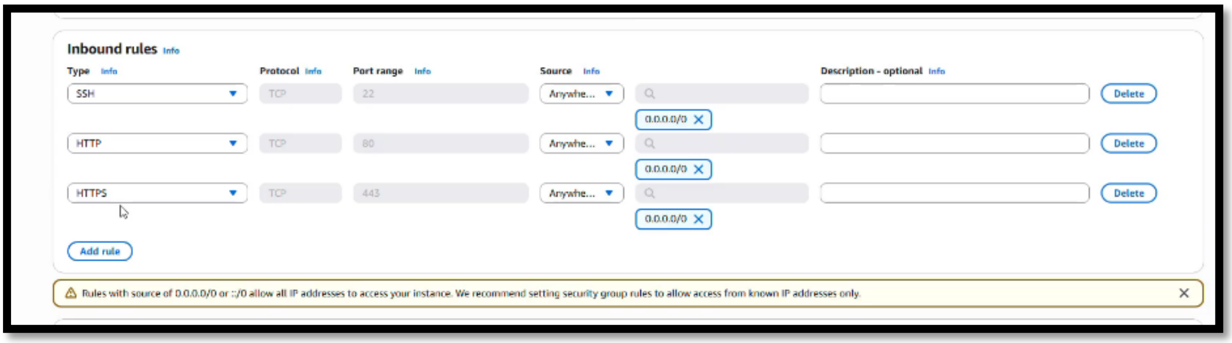


This is after you have done creating the VPC.

Creating EC2 Security Group

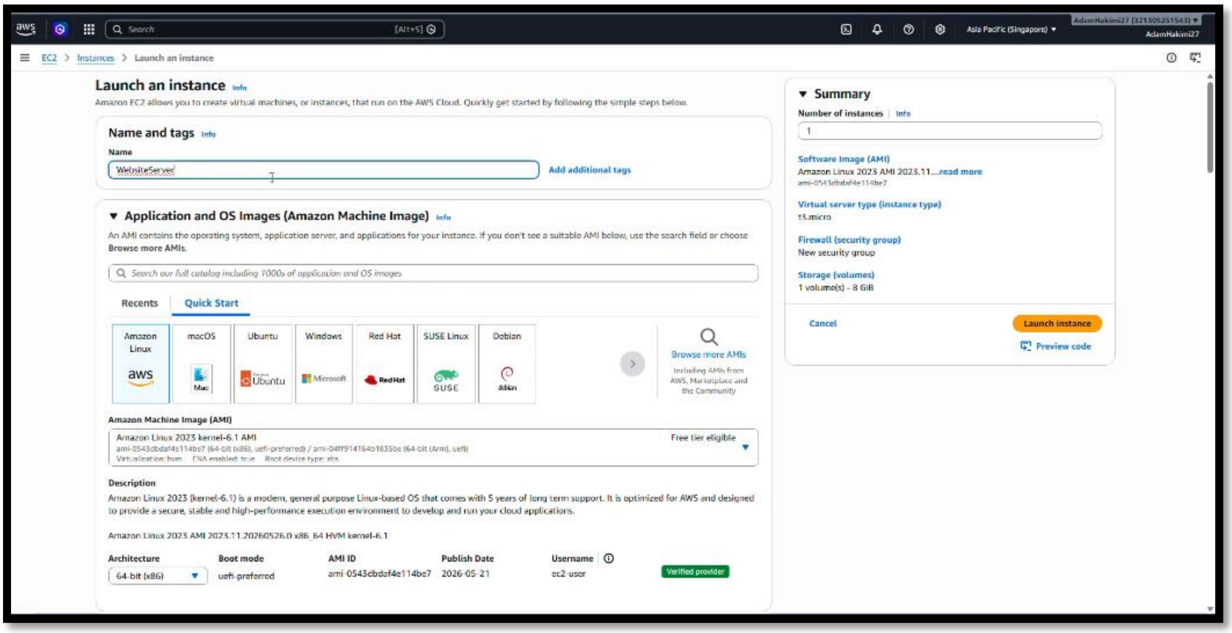


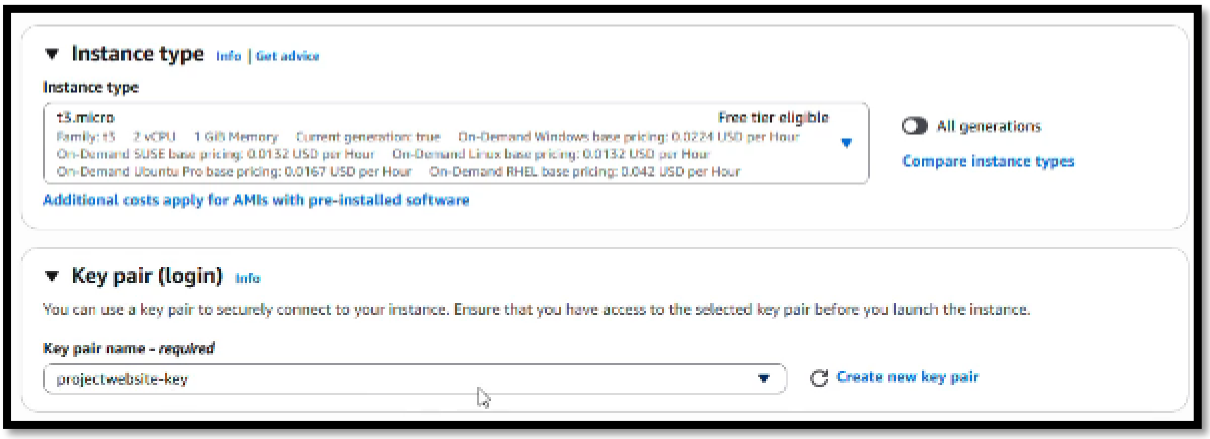
You can name the security group name based on you project to make a different from other security group name.

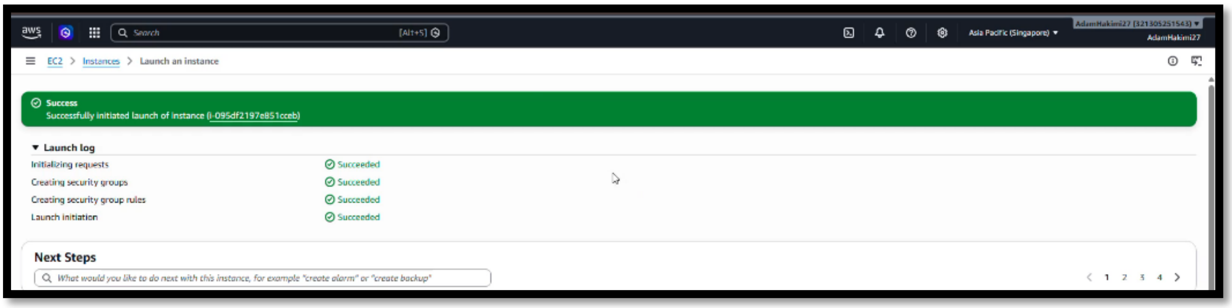


Three inbound rules were configured in the Security Group to support the functionality of the web server. The SSH rule (Port 22) allows administrators to remotely access the EC2 instance for configuration and maintenance purposes. The HTTP rule (Port 80) enables users to access the website through a web browser. The HTTPS rules (Port 443) provides secure encrypted communication between the website and users, ensuring data confidentiality and security. Together, these rules allow server management and web access while supporting secure communication.

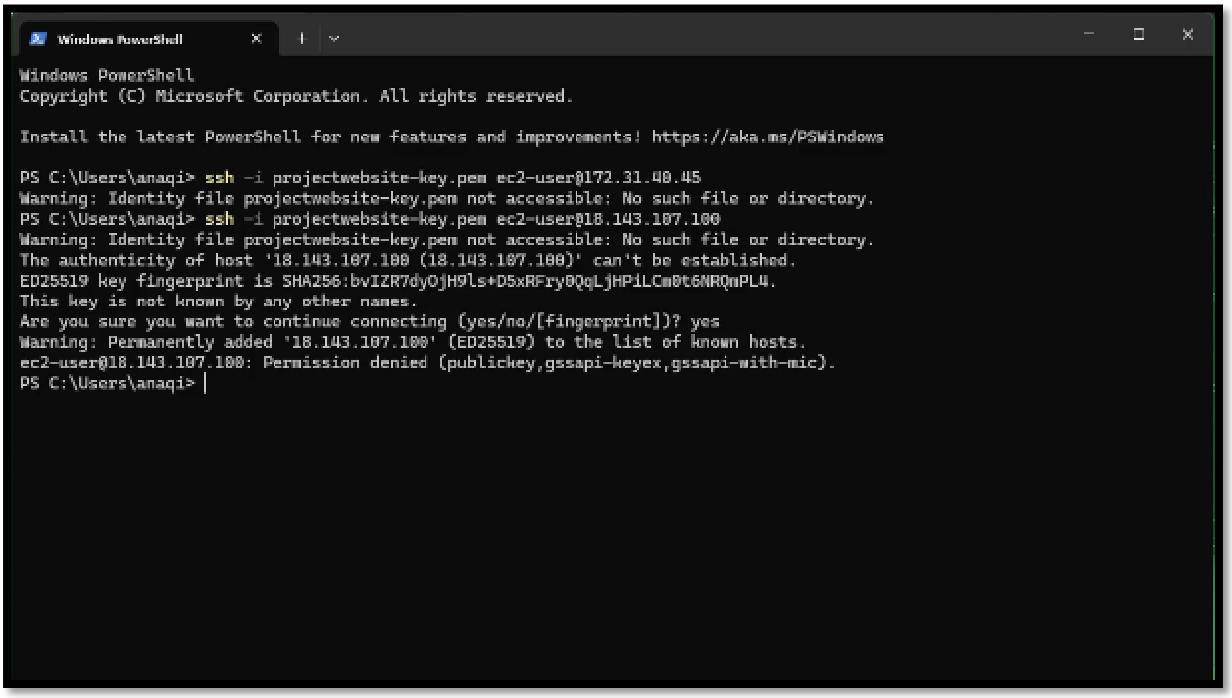
EC2 Launch Instance



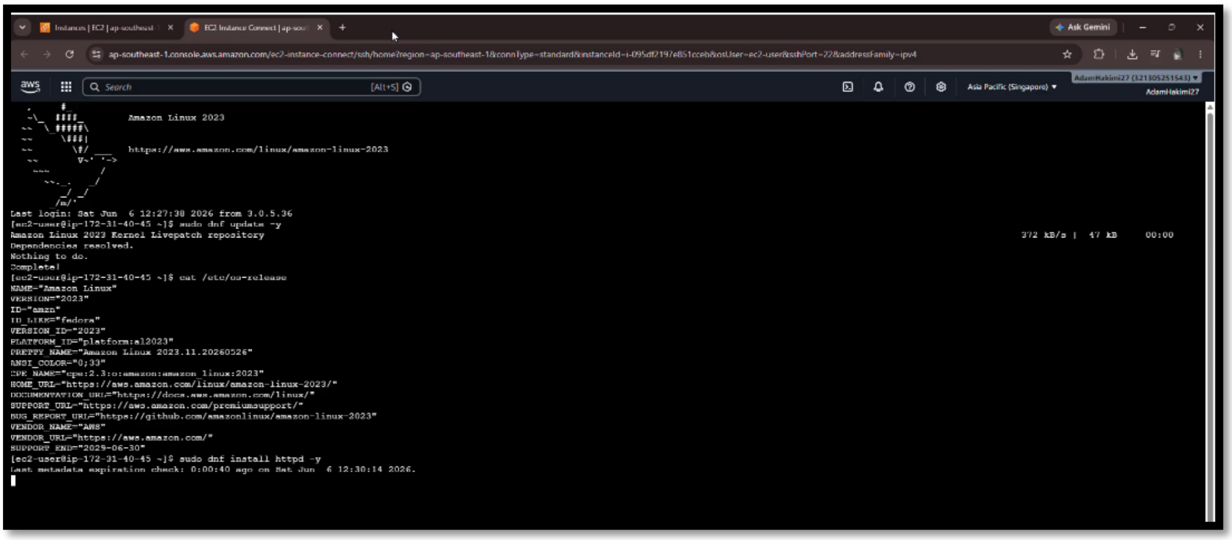




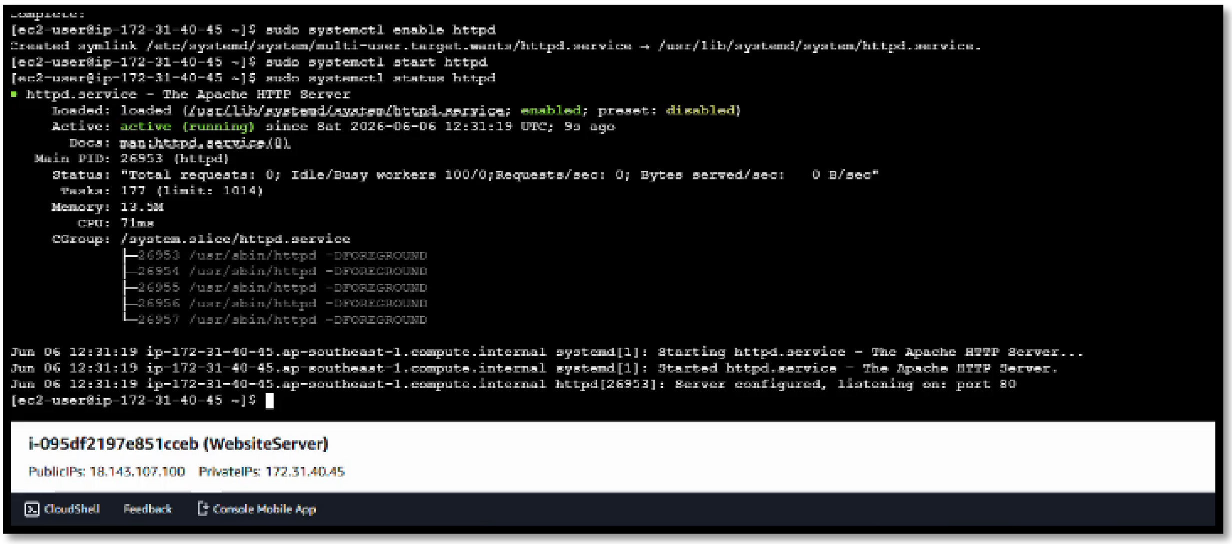
Connect to EC2 Instance



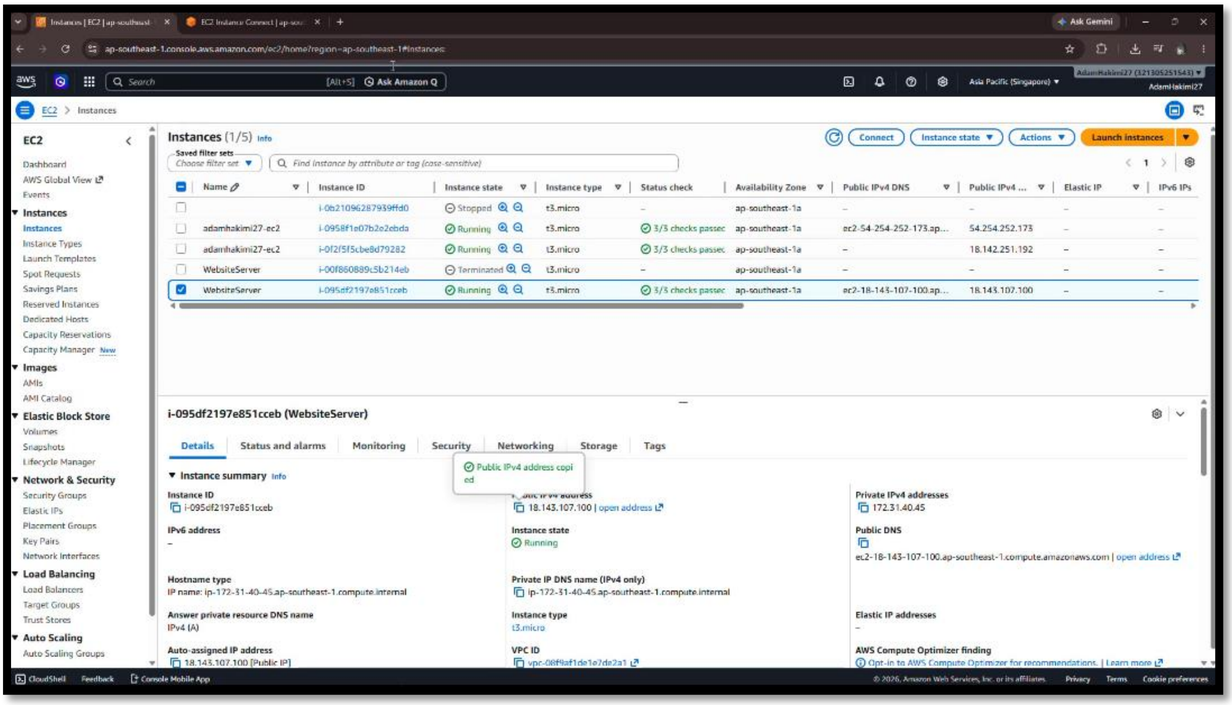
SSH was used to establish a secure connection between the local computer and the EC2 instance for server configuration and management.



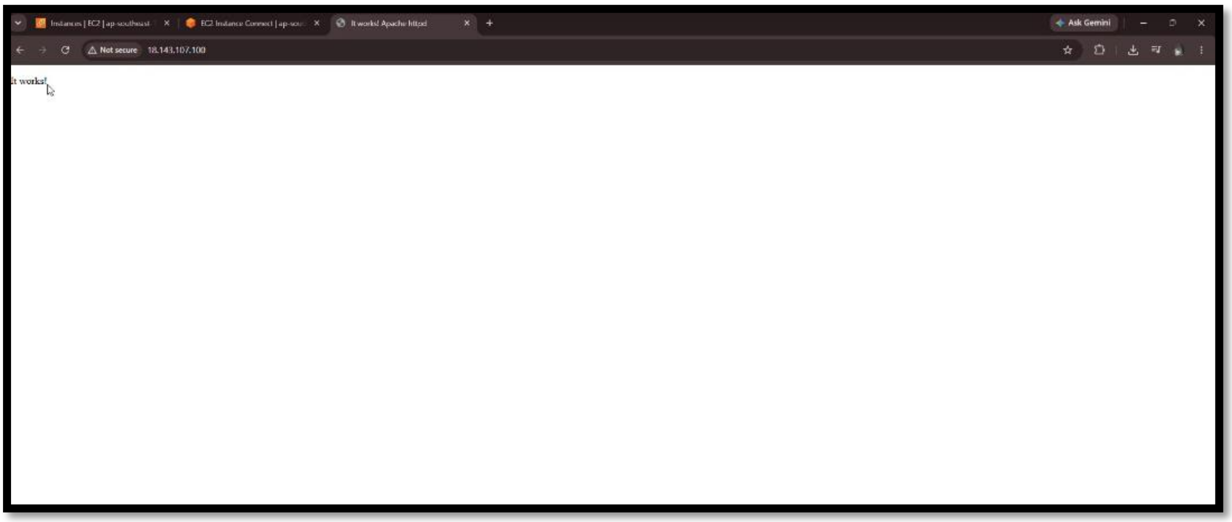
Apache web server was installed on the EC2 instance to enable website hosting.



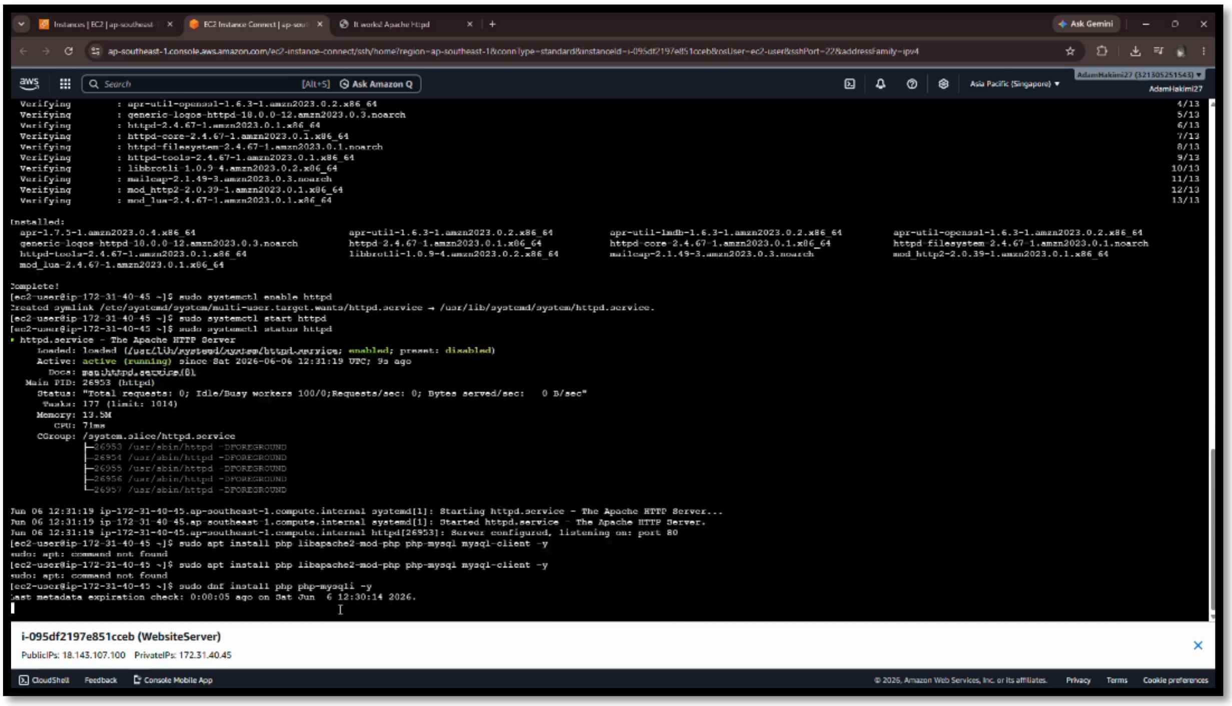
Apache was successfully enabled and started on the EC2 instance, and the service status confirmed that the web server was running correctly.



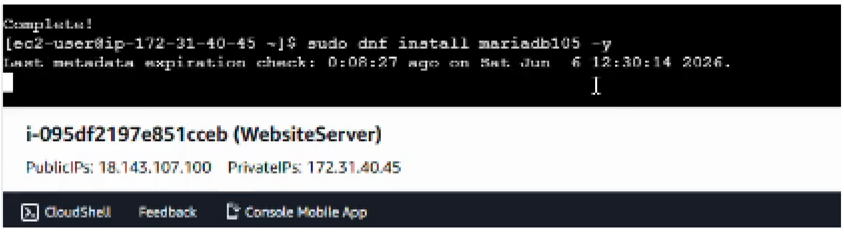
The EC2 instance was successfully deployed and assigned a public address for internet access.



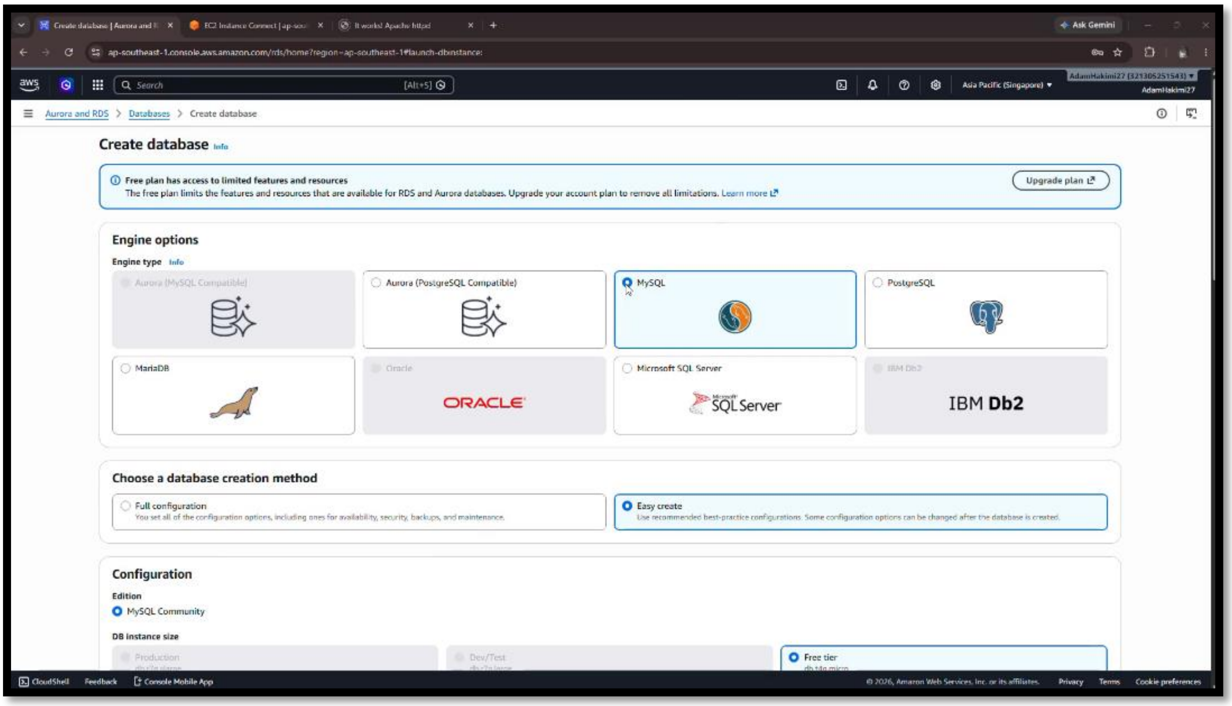
The Apache web server was successfully configured on the EC2 instance. Accessing the server’s public IP address displayed the default “It works!” page, confirming that the web server was operating correctly.

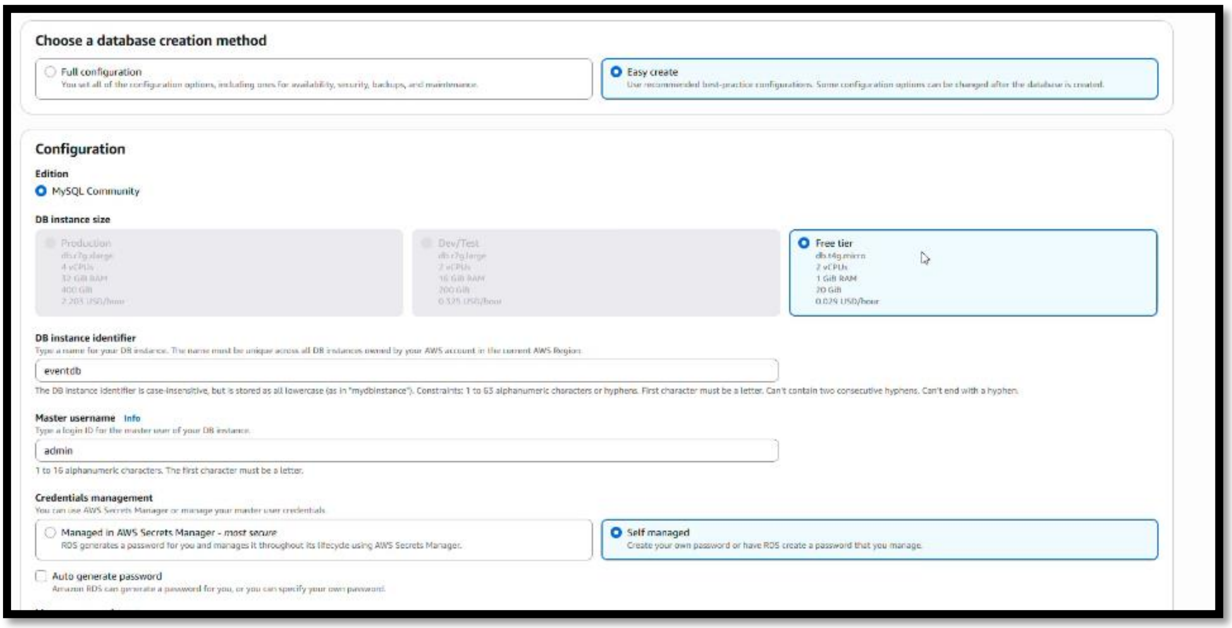


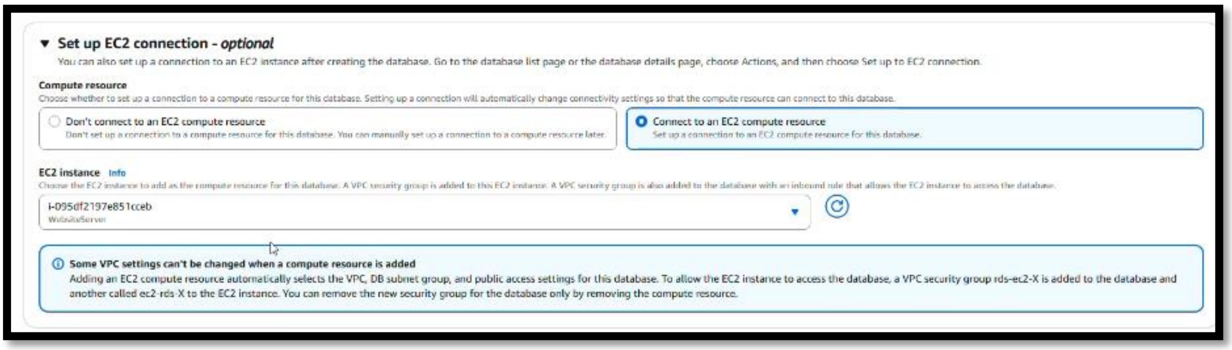
Installation of PHP and MySQL packages on the EC2 instance.



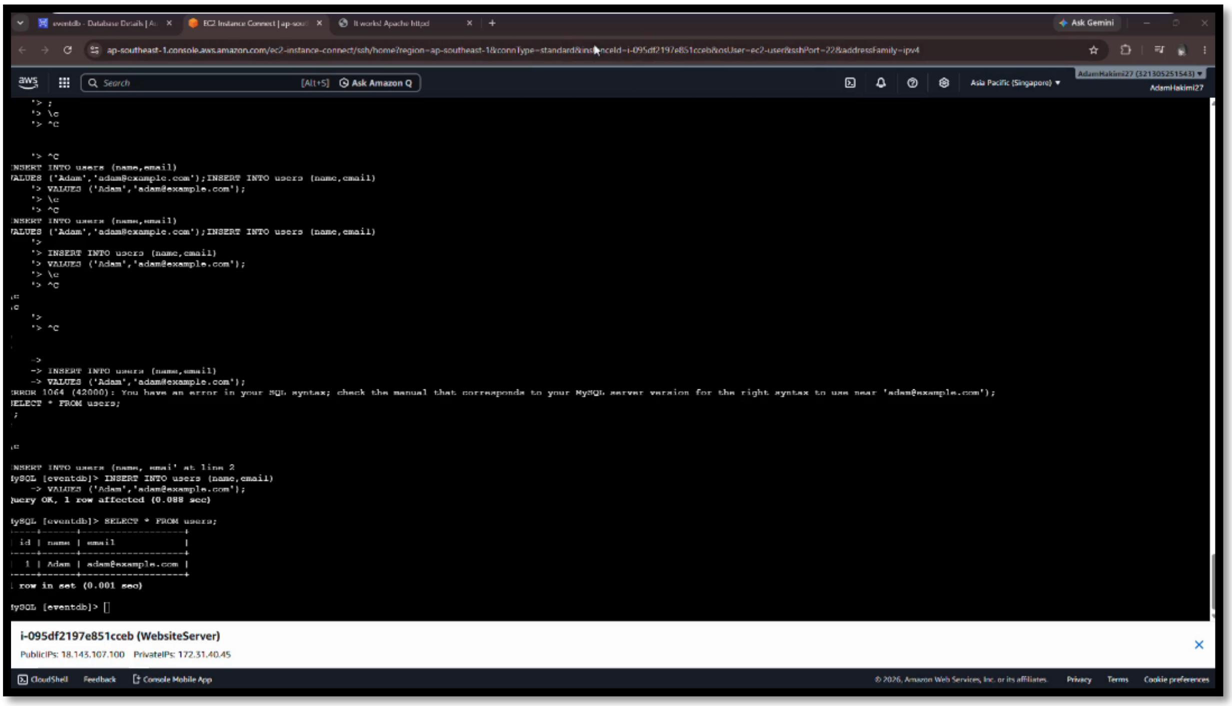
Installation of the MariaDB client on the EC2 instance.


Create database | Aurora and RDS
EC2 Instance Connect | ap-sout
It works! Apache httpd
Ask Gemini
ap-southeast-1.console.aws.amazon.com/rds/home?region=ap-southeast-1#launch-dbinstance:
Search
[Alt+S]
Asia Pacific (Singapore)
AdamHakimi27
Aurora and RDS > Databases > Create database
Create database Info
Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. Learn more
Upgrade plan
Engine options
Engine type Info
Aurora (MySQL Compatible)
Aurora (PostgreSQL Compatible)
MySQL
PostgreSQL
MariaDB
Oracle
ORACLE
Microsoft SQL Server
SQL Server
IBM Db2
IBM Db2
Choose a database creation method
Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Configuration
Edition
MySQL Community
DB instance size
Production
Dev/Test
Free tier
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

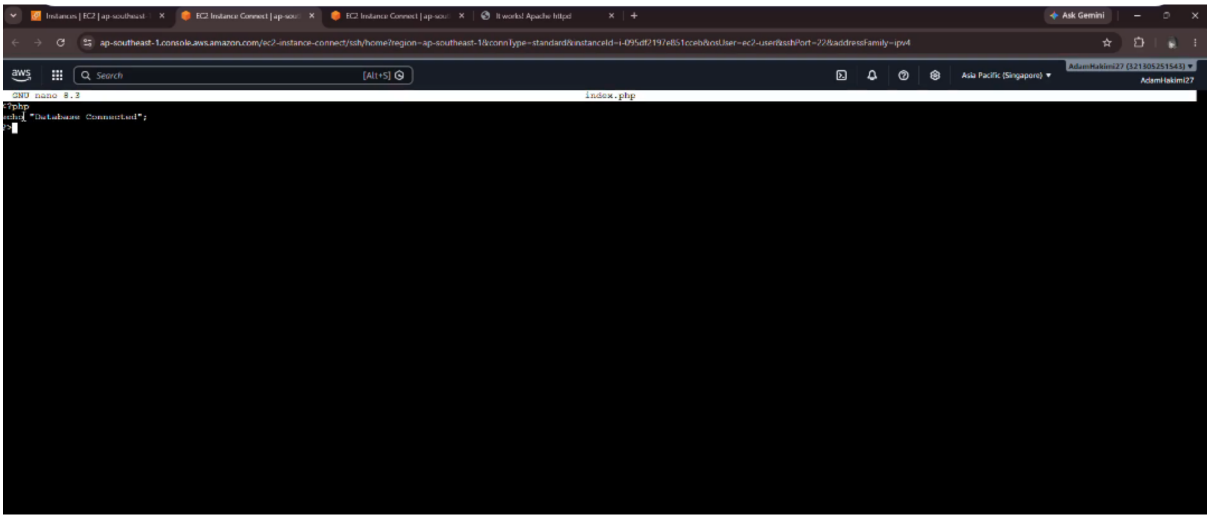

Choose a database creation method
Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Configuration
Edition
MySQL Community
DB instance size
Production
Dev/Test
Free tier
db.t4g.micro
2 vCPUs
1 GiB RAM
20 GiB
0.025 USD/hour
DB instance identifier
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
eventdb
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.
Master username Info
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.
Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
Self managed
Create your own password or have RDS create a password that you manage.
Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

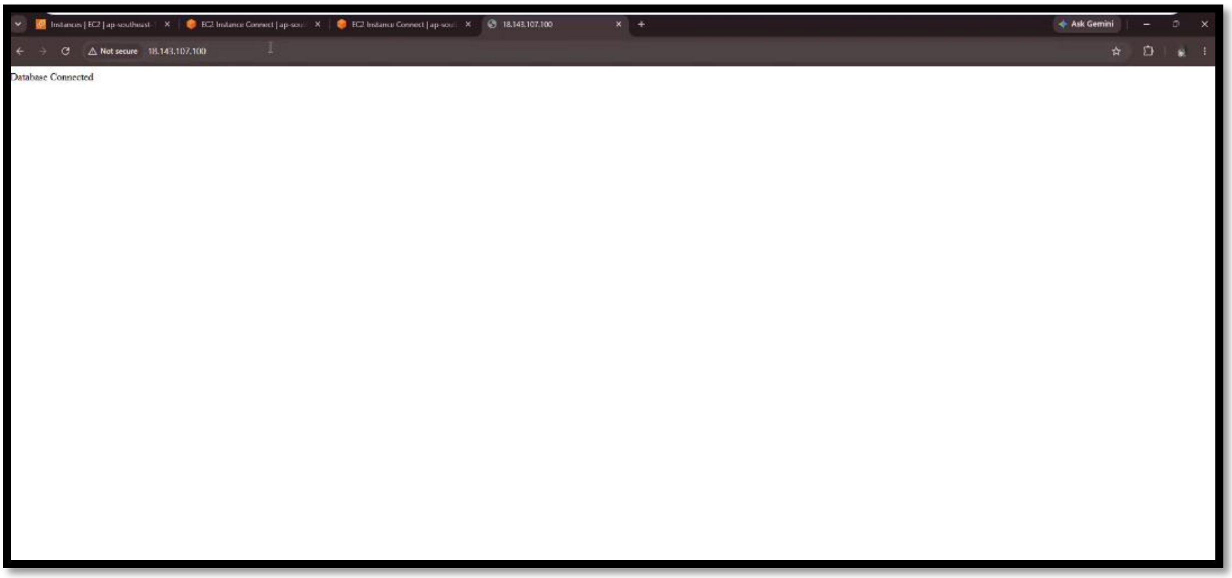

Set up EC2 connection - optional
You can also set up a connection to an EC2 instance after creating the database. Go to the database list page or the database details page, choose Actions, and then choose Set up to EC2 connection.
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
EC2 instance Info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.
i-095df2197e851cceb
WebsiteServer
Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Create Database

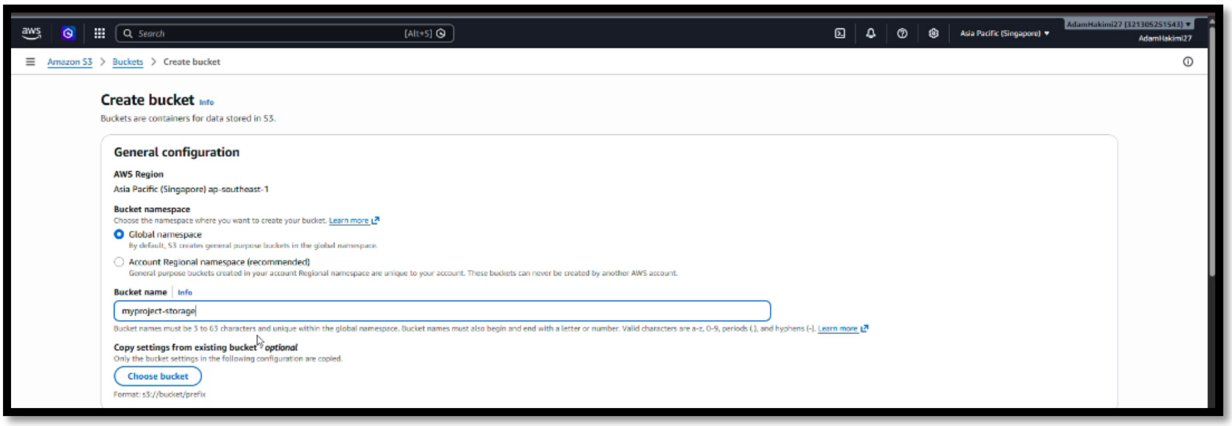


Testing Database connection

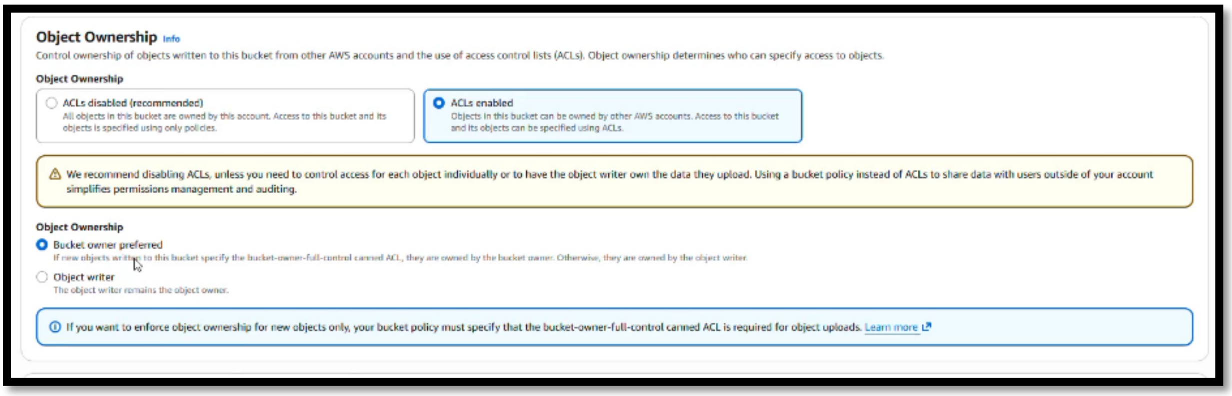


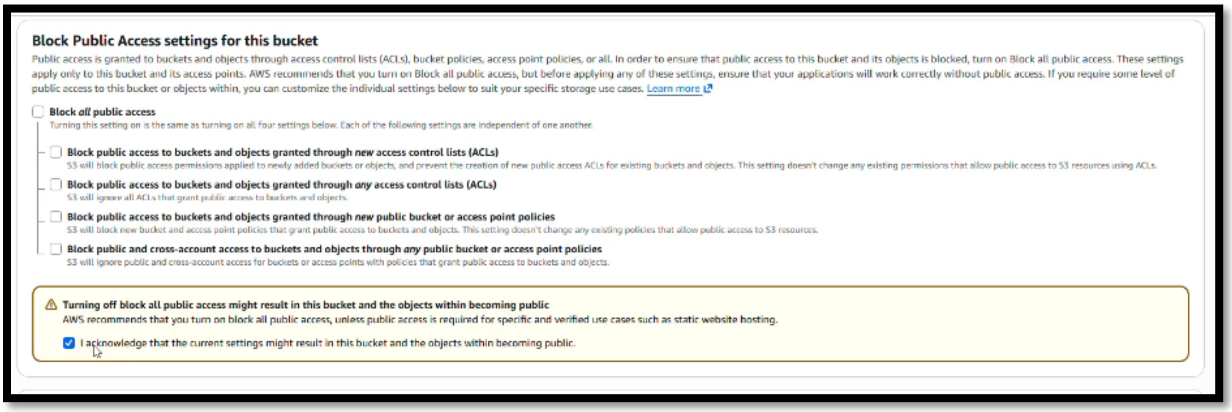


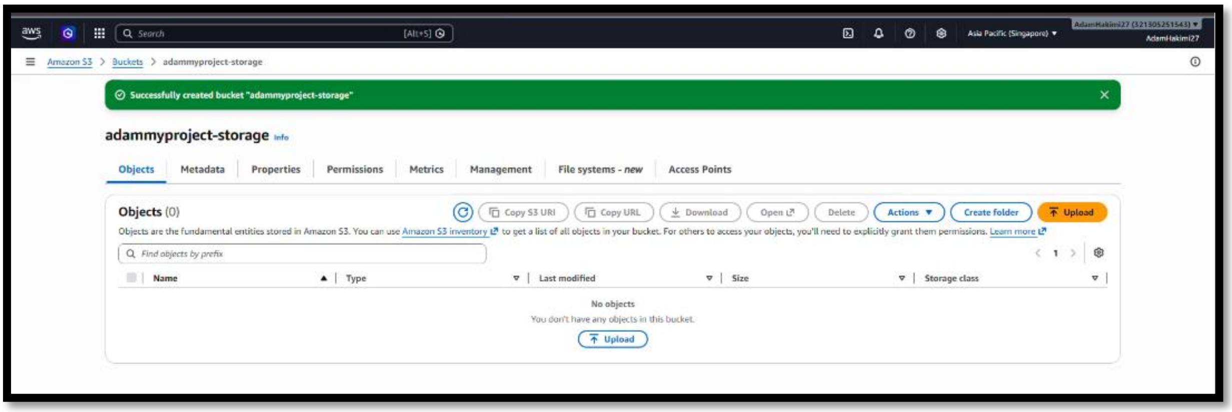
Using the same public IP and test the connection to see the database connection.

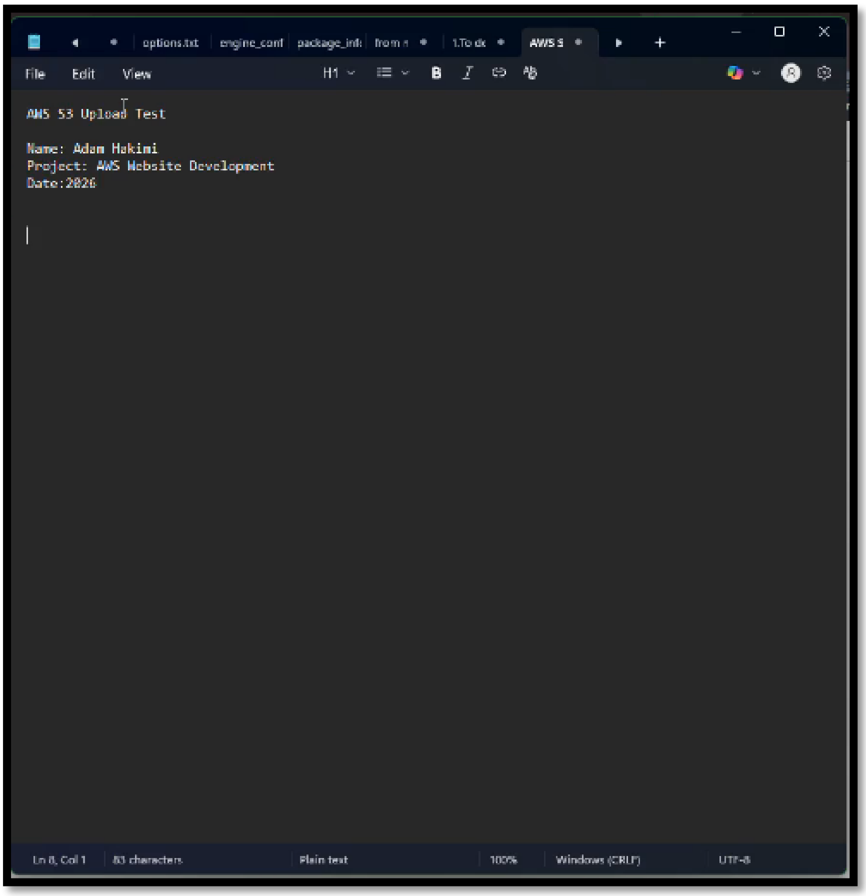


Creating Bucket

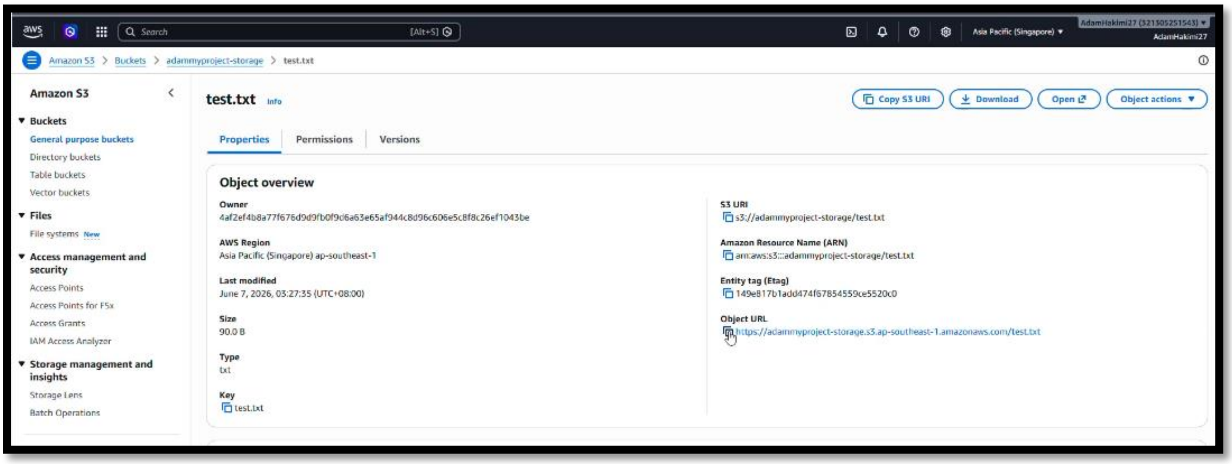




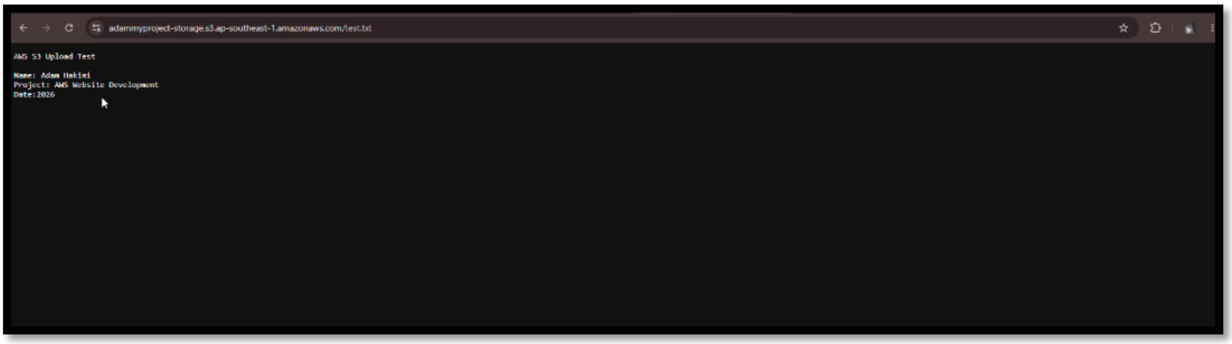


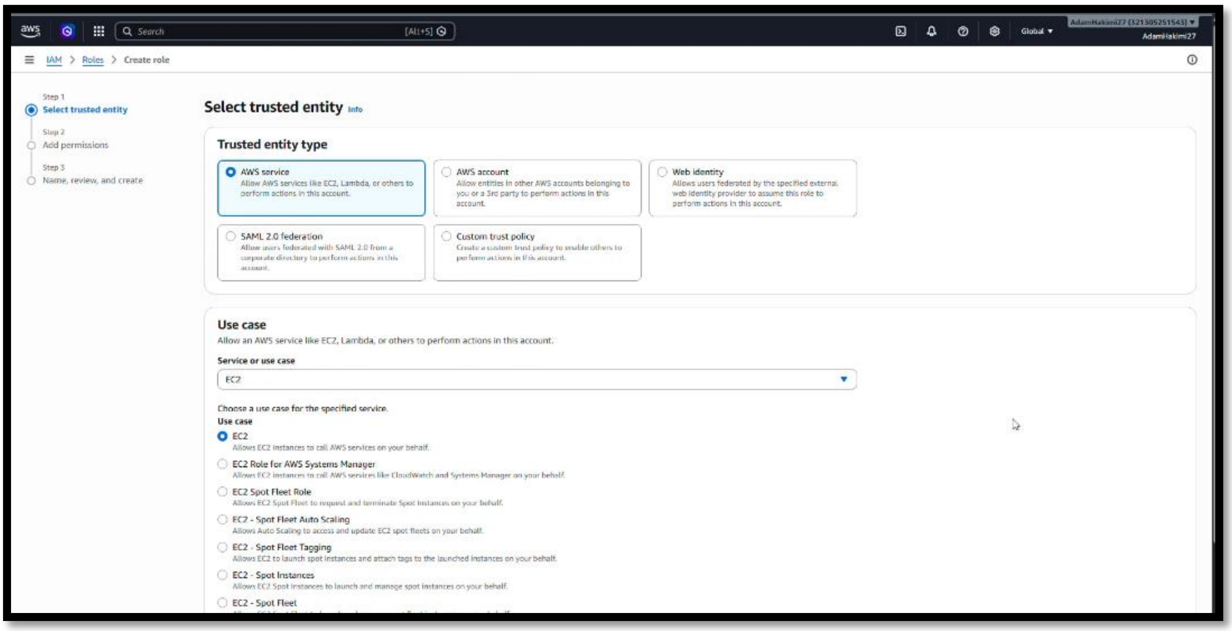


Creating txt file. For S3 upload test. Upload this txt file in the object.



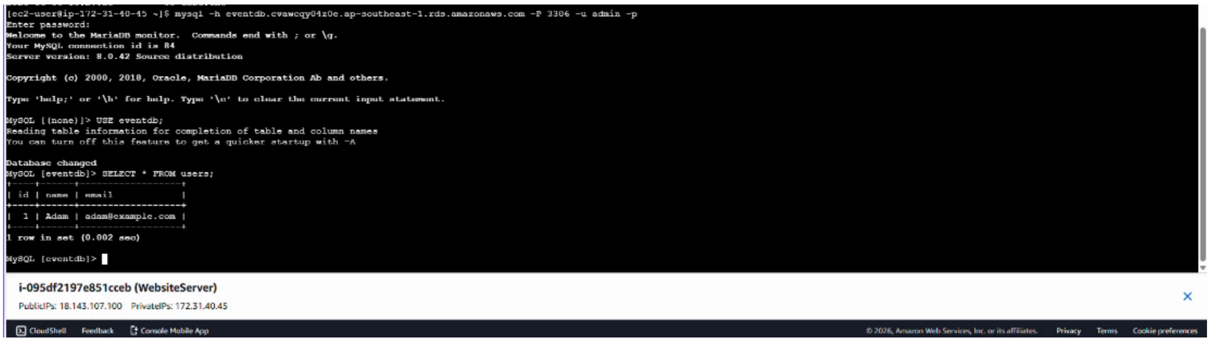
Use the object URL.



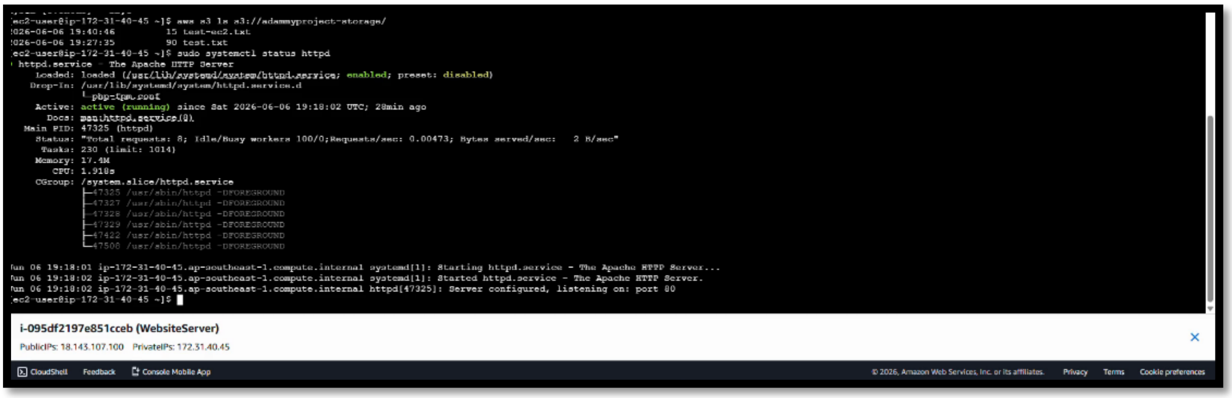


Creating IAM role.

A test file was successfully uploaded from the EC2 instance to the Amazon S3 bucket, confirming that EC2 could access and interact with S3.



The EC2 instance successfully connected to the RDS database and data was retrieved from the users table, confirming successful database integration.



The Apache web server was verified to be running successfully on the EC2 instance.

Summary

This project successfully demonstrated the deployment of a cloud-based web application using Amazon Web Services (AWS). The infrastructure was built using Amazon EC2 as the web server, Amazon RDS MySQL as the database service, and Amazon S3 as cloud storage. A security group was configured to allow SSH, HTTP, and HTTPS traffic for secure server management and web access.

The Apache web server, PHP, and MariaDB client were installed and configured on the EC2 instance. The website successfully hosted and verified through the EC2 public IP address. An Amazon RDS MySQL database was created, connected to the EC2 instance and tested by creating tables and inserting sample data.

Amazon S3 was integrated into the project to provide cloud storage functionality. Files were uploaded both manually and through the AWS CLI from the EC2 instance, confirming successful communication between EC2 and S3. Finally, all components were tested, including website accessibility, database connectivity, Apache service status, and S3 operations. The successful completion of these tests demonstrated that the cloud infrastructure was fully functional and met project objectives.